

Interview Answers in English

영어 면접 질문 대비 · Korean question labels in gray · [] = fill in with facts you have actually confirmed · yellow = most important

Speaking tips

- Short sentences. Pause between ideas. Speak slowly, you don't need a perfect accent.
- Don't memorize word for word. Remember the first sentence + 3 keywords.
- Useful phrases: *"Could you please repeat the question?"* / *"Let me organize my thoughts for a moment."* / *"I haven't confirmed that yet, but my understanding is ..."* / *"I'd like to discuss that with you if I'm admitted."*
- If the professor asks in Korean, answer in Korean. If he asks in English, answer in English. You can say a technical term in English inside a Korean sentence.

A. Motivation & Background

A1. Please introduce yourself and tell us why you applied. (자기소개와 지원동기)

Hello. After graduating from high school, I worked for about seven years in finance and marketing, and then I entered university at a later age. In an introductory biology class, I was fascinated by how precise and elegant living systems are, so I double-majored in life science. Later, I joined a wildfire analysis paper as a second author at the Nature and Symbiosis Research Institute, and that's when I realized how much I enjoy analyzing data. So I wanted to use these skills for genomic research on endangered species. Your lab takes part in the Vertebrate Genomes Project and has analyzed the genomes of many different species, so I felt it matches what I want to do very well.

A2. Why our lab? (왜 우리 연구실인가요)

As I understand it, your lab takes part in the Vertebrate Genomes Project and has studied the origin of African cattle and the genome of *chojongyong*, a rare parasitic plant on Dokdo. What attracted me most is that your lab has worked directly on the genomes of endangered and rare organisms. I want to read the genomes of endangered species and examine genetic diversity and inbreeding, so I want to learn from a lab that has done this kind of work. I also haven't run analyses on a server yet, and I'd like to fill that gap by following the lab's real analysis workflow.

A3. Why change from business to life science? (7년 일하다 생명과학?)

It wasn't a vague dream. I tested it. I studied life science as a double major, joined citizen-science projects such as forest insect and otter surveys, and worked at a research institute and at the Korea National Arboretum. Even when it was hard, I felt I wanted to keep doing it. My marketing experience, such as setting KPIs and reading data, also helps me interpret data now.

A4. Why genomics rather than ecology or forestry? (왜 유전체인가)

I love life science itself. From the moment of fertilization to how an organism keeps itself alive, it's amazing and elegant, and the more I learn, the more fascinating it gets, even though it is difficult. I also have a special feeling for endangered species. I realized the field that combines both interests, the microscopic world and endangered species, is conservation genomics. I thought it's a field where I can dig deep with passion for a long time. I'll keep an ecological perspective when I connect results to the field.

A5. Master's only, or a PhD too? (석사만? 박사까지?)

I'm seriously considering a PhD. I see the master's program as the time to build a foundation, and I think a PhD is necessary to become an independent researcher. So if I'm admitted, I want to build my skills first and then continue to a PhD. Of course, I'll do my best in the master's program itself.

B. Experience & Skills

B1. What exactly did you do in your wildfire paper? (산불 논문)

I participated as the second author. I was in charge of designing the research approach, collecting and preprocessing the data, running the analysis, writing the first draft, and making the figures. The outcome was whether a wildfire occurred. As explanatory variables, I chose human factors such as forest roads, hiking trails, and forest management, together with forest-stand variables and weather variables. First, I ran Random Forest and XGBoost to see which variables were most strongly related to wildfire occurrence. Then I removed variables with high VIF scores, meaning they overlapped too much with other variables, and ran a logistic regression with the final set of variables. The results showed that fewer wildfires occurred in older stands and at higher humidity, and more wildfires occurred in coniferous forests. Wildfires tended to be suppressed closer to forest roads and trails. Forest management variables such as thinning were not statistically significant, so further research is needed. What impressed me most was that older stand age was associated with fewer wildfires.

B1-1. What is a Random Forest? (랜덤 포레스트)

It builds many decision trees and combines their results. Each tree uses slightly different data and variables, so even if one tree is wrong, the whole model stays stable. Also, by checking how much each variable improves the prediction, I can see variable importance. I used it to find which variables were strongly related to wildfires.

B1-2. How is XGBoost different? (XGBoost와 차이)

Random Forest builds trees independently and combines them. XGBoost builds trees one after another, and each new tree corrects the errors of the previous ones. It usually performs better, but it is sensitive to settings, so I have to be careful about overfitting.

B1-3. What is VIF, and why did you remove variables? (VIF)

VIF, or variance inflation factor, shows how much a variable overlaps with the others. If a variable can be explained by the other variables, its VIF is high. A VIF above 5 or 10 is usually considered a problem. Overlapping variables make regression coefficients unstable and hard to interpret, so I removed the high-VIF variables.

B1-4. Why did you run logistic regression after machine learning? (왜 로지스틱 회귀?)

Machine learning tells me which variables are important, but it's hard to read the direction and size of the effect directly. So after selecting the important variables, I used logistic regression to interpret whether each variable increases or decreases wildfire risk. [Check that this matches the real reason in your paper.]

B1-5. Why would wildfires be suppressed closer to forest roads? (임도 가까울수록 억제, 왜?)

As I understand it, when there is a forest road, firefighters and equipment can reach the site quickly, so they can put the fire out sooner and the fire doesn't spread. But I see this as an association rather than proof of causation, and I think further research is needed.

B1-6. Why were fewer wildfires found in older stands? What are the limitations? (임령, 한계)

In our paper, we interpreted it as [the explanation in your paper's discussion]. As limitations, we looked only at whether a fire occurred, not its size or cause, and the forest management variables such as thinning were not significant, so further research is needed.

B2. Do you have experience with Linux or servers? (서버·리눅스 경험)

To be honest, I haven't run analyses on a Linux server yet. I have used Python for machine learning and visualization, and GIS for spatial data. I'm currently studying Linux on my own to learn the basics. If I'm admitted, I'll follow the lab's pipeline and learn from it, and my goal in the first one or two months is to run one public dataset from start to finish and reproduce the results.

B3. You have many activities. Can you focus on research? (활동이 많은데?)

All of my activities were ultimately about biological conservation. They came from my wish to protect biodiversity, so they point in the same direction as my research. I also believe my love for living things will keep me from burning out. Stamina matters in research too, so I'll sometimes go to forests and nature to meet the living things we share the world with, and recharge. I'll put research first and manage my time so that these things never get in the way.

B4. You're older and have a short research career. Can you keep up? (나이, 짧은 연구 경력)

Yes, I'm older, but I believe learning is something we continue for life. I started later than others, but I have an objective view of what I like, what I'm good at, and what I lack. I'll make up for what I lack, and use the know-how from my varied experiences to become a good researcher.

C. Research Plan (most important)

C1. Please describe your research plan in about two minutes. (연구계획 2분)

I'd like to re-analyze the genome of the long-tailed goral, to evaluate its inbreeding and genetic diversity. The goral is an endangered species in Korea, and there is concern about population decline from habitat fragmentation and heavy snow. In 2018, a previous study sequenced one individual at 27x coverage and analyzed it by mapping to the sheep genome. But sheep split from the goral lineage about ten million years ago, so the sequence difference to the reference is roughly 2%, while heterozygosity is only about 0.1%. That means the reference can affect the results. In 2024, a chromosome-level reference genome of the Himalayan goral, in the same genus, was published, so now I can remap the same public data to a much closer reference.

For the method, I'll map the data to the new reference and call variants, and then look at three things. Heterozygosity shows genetic diversity, ROH, or runs of homozygosity, shows the degree of inbreeding, and PSMC shows changes in past population size. I'll compare the results from the sheep reference and the new reference under the same conditions, to see how much the reference changes the results. I know the public whole-genome data is a single individual, so I can't draw population-level conclusions. I'd like to discuss the details and how to add more individuals with you. The results could provide baseline data for conservation and restoration policy.

C2. That data was already analyzed in 2018. What's new? (새로움)

What's new is the analysis, not the data. First, I'll run the same data under the same conditions against two references, and compare heterozygosity and PSMC. That lets me directly check whether the previous result was biased by the reference. Second, I'll add ROH analysis, which the previous study didn't do, with chromosome positions, to evaluate inbreeding. Also, previous studies used different mutation rates and generation times for PSMC, so their results can't be compared directly. I can standardize these and compare.

C3. How will you check the change, and what counts as success? (확인 방법, 성공 기준)

I'll map the same data to both references under the same conditions, and compare the heterozygosity values and the PSMC curves. If they differ, the earlier result was affected by the reference. If they don't, it confirms that the earlier result is reliable. I think either outcome is meaningful.

C4. 94% of reads already mapped to sheep. Why do you need a closer reference? (94% 매핑인데?)

A 94% mapping rate means the reads were placed, but not necessarily placed correctly. The difference from sheep is about 2%, while heterozygosity is only about 0.1%, so even small misalignments can affect the estimate. I'll check how much it actually matters by comparing the two references.

C5. You only have one individual. What can you conclude? (개체가 1개)

You're right. With one individual, I can't draw population-level conclusions. So the scope of this study is the genomic state of that individual and a comparison of methods. I also know that Korean gorals are divided into northern and southern populations, so one individual cannot represent the whole. I'd like to discuss with you how to add more individuals, for example through archived samples or collaborating institutions. If that's hard, I'm also thinking of applying the same approach to another species, such as the otter.

C6. What if the public data is insufficient? (Plan B)

I'll apply PSMC only to individuals with enough coverage, which here is the 27x individual. For low-coverage data, I'll look for methods that work at low coverage, or use it only for population-level analysis. If the data is truly insufficient, I'll apply the same framework to another endangered species such as the otter.

C7. Does low heterozygosity always mean high extinction risk? (이형접합도 낮으면 위험?)

Not necessarily. A population that stayed small for a long time may have already purged harmful mutations. On the other hand, a population that declined recently may still have high heterozygosity but face rapidly increasing inbreeding. So I think ROH length and PSMC should be interpreted together, and I wouldn't conclude inbreeding from the heterozygosity of one individual alone.

C8. What is ROH, and how does PSMC work? (ROH, PSMC)

ROH is a long stretch where the two chromosomes carry identical alleles. Recombination breaks these stretches into shorter pieces over time, so long ROH suggests recent inbreeding and short ROH suggests more ancient ancestry. PSMC estimates past population size from the distribution of heterozygous sites in a single genome. Its limitations are that the time scale depends on the mutation rate and generation time, and that its resolution for recent time is low.

C9. How will you handle sex chromosomes and reference quality? (성염색체, 참조 품질)

The reference individual is female, so there is no Y chromosome, and the paper identified the X by comparing it with cattle and muskox. I plan to exclude the X and use autosomes only. The reference is chromosome-level, with a scaffold N50 of about 106 Mb and BUSCO completeness of about 97%. But the contig N50 is only about 3.7 Mb, so I'll be careful near contig boundaries.

C10. The reference is from China. How different is it from Korean gorals? (중국 개체 참조)

They are different species in the same genus, so there is some difference. But they are much closer than sheep. I'll check this with mapping rates and by comparing the two references. I haven't confirmed the exact divergence time yet.

C11. What if I suggest a different topic? (다른 주제 제안 시)

My plan is a draft, and your advice and the lab's direction come first. The workflow of mapping, variant calling, and population genetic metrics is shared across species, so I'll learn a lot from any topic you give me, and I can continue my own interests on top of that.

C12. You also like otters. Why the goral instead? (수달 vs 산양)

I'm interested in both. I met otters directly through citizen-science activities. I chose the goral first because public data and a closer reference are available, so I can start without new samples. For the otter, a same-species reference and whole-genome data of East Asian individuals are public, so if I can get Korean samples, I'd like to extend the work. I'd like to discuss which direction is better with you.

C13. How can the results be used for policy? Is it feasible in two years? (정책 활용, 2년 일정)

By showing inbreeding levels and population history with numbers, the results become baseline data for discussing habitat connectivity and which individuals to bring in for restoration. In the Worak Mountain restoration, which population to translocate from was an actual issue. For the schedule: in the first semester, coursework, learning the pipeline, and reviewing the literature. In the second, data processing and mapping. In the third, analysis and interpretation. In the fourth, thesis writing. If the scope is too big, I'll focus on one species and a small number of individuals.

Courses on the application (Genetics / Genomics / Ecology)

E1. What do you remember from your genetics course? (유전학)

The course covered the basics from Mendelian inheritance to gene expression and regulation. What stayed with me is how allele frequencies change over generations, and how inbreeding leaves genetic consequences in a population. Heterozygosity and ROH, which I'll use in my research, extend these ideas to the population level. Under inbreeding, the heterozygous proportion decreases to $2pq$ times $(1 - F)$.

E2. What did you learn in genomics? (유전체학)

I learned the workflow of sequencing, assembly, and annotation. Reads are assembled into contigs, then scaffolds, and then chromosomes, using methods such as Hi-C. Annotation then marks gene locations and functions. I also learned how sequencing data is generated and how to use public databases such as NCBI. In my research, I won't build a new assembly. I'll map reads to an existing reference and call variants.

E3. What did you learn in ecology, and how does it connect? (생태학)

The course covered how organisms interact with their environment at the population, community, and ecosystem levels. I learned how habitat fragmentation affects population size and how biodiversity is assessed. To connect genomic results to conservation in the field, I need this ecological perspective. When habitats fragment, gene flow decreases, and small populations lose diversity through drift and inbreeding.

D. Other

D1. Which recent issues in your field have you followed? (최신 이슈)

Honestly, I can't follow everything, but there are two things I understand. First, chromosome-level reference genomes keep being published, and my research plan started from that. Second, in conservation genomics, researchers don't look only at heterozygosity, but also at genetic load and genetic rescue. For your lab's papers, I've read the paper on the rare parasitic plant on Dokdo, and I think I need to read more. If I'm admitted, I'll study more deeply.

D2. Do you have anything to add, or any questions? (마지막 질문)

Is there anything I should prepare before admission? I'm also wondering whether a conservation genomics project based on public data could work as a master's topic. Thank you for your time today.

Explain it in one sentence (영어로 한 줄 설명)

Term	One-sentence explanation
Reference genome	A representative genome sequence that we align new reads to, like a map.
Resequencing	Sequencing an individual of a species that already has a reference genome and comparing it to that reference.
Sequencing depth (coverage)	The average number of times each position in the genome is read; 27x means each position is read about 27 times.
Variant calling	Finding positions where an individual's genome differs from the reference.
Heterozygosity	The proportion of positions where the two inherited alleles are different; it reflects genetic diversity.
Runs of homozygosity (ROH)	Long stretches where both chromosomes carry the same alleles; long ROH suggests recent inbreeding.
PSMC	A method that estimates past population size from the pattern of heterozygous sites in one genome.
Effective population size (N_e)	The size of an idealized population that would show the same amount of genetic drift as the real one.
Reference bias	When reads that differ a lot from the reference map poorly or incorrectly, so heterozygosity and variants are distorted.
Inbreeding depression	Reduced survival and reproduction because harmful recessive alleles become homozygous.
Genetic load	The burden that harmful mutations place on a population's fitness.
Genetic rescue	Bringing in individuals from another population to restore genetic diversity.
Habitat fragmentation	When habitat is broken into isolated patches, reducing movement and gene flow.
N50	A measure of assembly contiguity; half of the assembly is in pieces at least this long.
Random Forest	An ensemble of many decision trees built independently, whose results are combined.
XGBoost	A gradient boosting method that builds trees sequentially, each correcting the errors of the previous ones.
VIF	A measure of how much a variable is explained by the other variables; high values mean multicollinearity.
Logistic regression	A regression for yes/no outcomes that gives the direction and size of each variable's effect, such as odds ratios.